

Nathaniel Pritchard

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CONTACT	Primary Email: pritchard@maths.ox.ac.uk Website: https://nathanielpritchard.github.io
RESEARCH INTERESTS	Generalized Linear Models. High Performance Computing. Bayesian Statistics. Iterative Solvers. Optimization. Preconditioners. Random Sketching. Statistical Computation.
EDUCATION	University of Wisconsin - Madison, Ph.D., Statistics Adviser: Vivak Patel Madison, WI August 2018 - May 2024 University of North Carolina at Chapel Hill B.S., Statistics and Analytics (Highest Honors and Highest Distinction) Honors Thesis Adviser: Shankar Bhamidi Chapel Hill, NC August 2014 - May 2018
RESEARCH EMPLOYMENT	University of Oxford, Post Doctoral Research Associate Supervisor: Yuji Nakasukasa and Per-Gunnar Martinsson Topic: Certifying accuracy of randomised algorithms in numerical linear algebra Oxford, United Kingdom October 2024 - University of Wisconsin-Madison, Research Assistant Supervisor: Vivak Patel Topic: Development of Randomized Linear Algebra Library for Julia Madison, Wisconsin August 2023 - May 2024 Argonne National Labs, Givens Associate Supervisor: Adrian Maldonado Topic: Accelerating Newton-Raphson on GPUs using deflation methods Chicago, Illinois May 2023 - July 2023 Argonne National Labs, Givens Associate Supervisor: Adrian Maldonado Topic: Preconditioners for solving graph Laplacians arising from power grid networks Chicago, Illinois May 2020 - July 2020
PRE-PRINTS	Patel, V., Maldonado, D. A., Melnichenko, M., Pritchard, N. , Rao, V., Rebrova, E., & Sankararaman, S. (2025). Scientific Applications Leveraging Randomized Linear Algebra. arXiv preprint arXiv:2506.16457.

- PUBLICATIONS** **Pritchard, N.**, & Patel, V. (2023). Solving, Tracking and Stopping Streaming Linear Inverse Problems. *Inverse Problems*, 40 (8), 10.1088/1361-6420/ad5583.
- Pritchard, N.**, & Patel, V. (2023). Towards Practical Large-Scale Randomized Iterative Least Squares Solvers through Uncertainty Quantification. *SIAM/ASA Journal on Uncertainty Quantification*, 11(3), 10.1137/22M1515057.
- He, M., Glasser, J., **Pritchard, N.**, Bhamidi, S., & Kaza, N. (2020). Demarcating geographic regions using community detection in commuting networks with significant self-loops. *PloS one* 15(4), e0230941.
- TALKS** “Small Sketches for Big Matrix Approximations” July 2025 SIAM Annual Meeting in Montreal, Canada.
- “Small Sketches for Big Matrix Approximations” June 2025 at The 30th Biennial Numerical Analysis Conference in Glasgow, Scotland.
- “Statistical Approaches to Numerical Linear Algebra” March 2025 at Optimization and Numerical Linear Algebra Exposé at the Rutherford Appleton Laboratory in Didcot, UK.
- “Who needs a residual when an approximation will do?” December 2024 at Numerical Analysis Internal Seminar in Oxford, UK.
- “Tracking Progress in a Big Data World“ October 2024 at SIAM Mathematics of Data Science in Atlanta, Georgia.
- “Iterative Right Random Sketching for Large-Scale Logistic Regression.“ August 2024 at Joint Statistical Meeting in Portland, Oregon.
- “Computationally Efficient Tracking for Iterative Random Sketching.“ May 2024 at SIAM Applied Linear Algebra Conference in Paris, France.
- “Computationally Efficient Tracking for Iterative Random Sketching.“ April 2024 at IFDS Ideas Forum in Madison, Wisconsin.
- “Large-scale randomized iterative least squares.” March 2023 at SIAM CSE in Amsterdam, Netherlands.
- “Towards practical large-scale least squares solvers with Iterative Right Random Sketching.” February 2023 at Argonne National Labs LANS seminar in Chicago, Illinois (Virtual).
- “Residual Tracking and Stopping for Iterative Random Sketching” April 2022 at Copper Mountain Conference on Iterative Methods in Copper Mountain, Colorado (Virtual).

HONORS & AWARDS	SIAM Student Travel Award (SIAM LA 2024)	May 2024.
	Outstanding TA	May 2023.
	Student Research Grant Competition UW - Madison	February 2023.
	Outstanding TA (Honorable Mention)	May 2022.
SERVICE	Committee Service: TA Training Redesign (2020), Space Committee (2021), Awards and Outreach Committee (2022).	
	Statistics Graduate Student Association: Outreach Chair (Spring 2018 - Fall 2020), President (Fall 2020 - Spring 2023).	
	Journal Reviewer: IMA Journal of Numerical Analysis	
SKILLS	Computer Languages: Julia, R, C, C++, Python	
	APIs: CUDA, MPI, OpenMP, PETSc	
SOFTWARE	RLinearAlgebra (Co - Developer): A Julia Package for prototyping and benchmarking Randomized Linear Algebra algorithms.	